جامعة جدة
University of Jeddah

College of Computer Science and Engineering
CCSW-313 SE1

RFID/ Fingerprint/ Camera based smart student and employee attendance system

Student Name	Student ID
Raghad Mohammed Alsaiari	2210207
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deema ibrahim alsini	2215727

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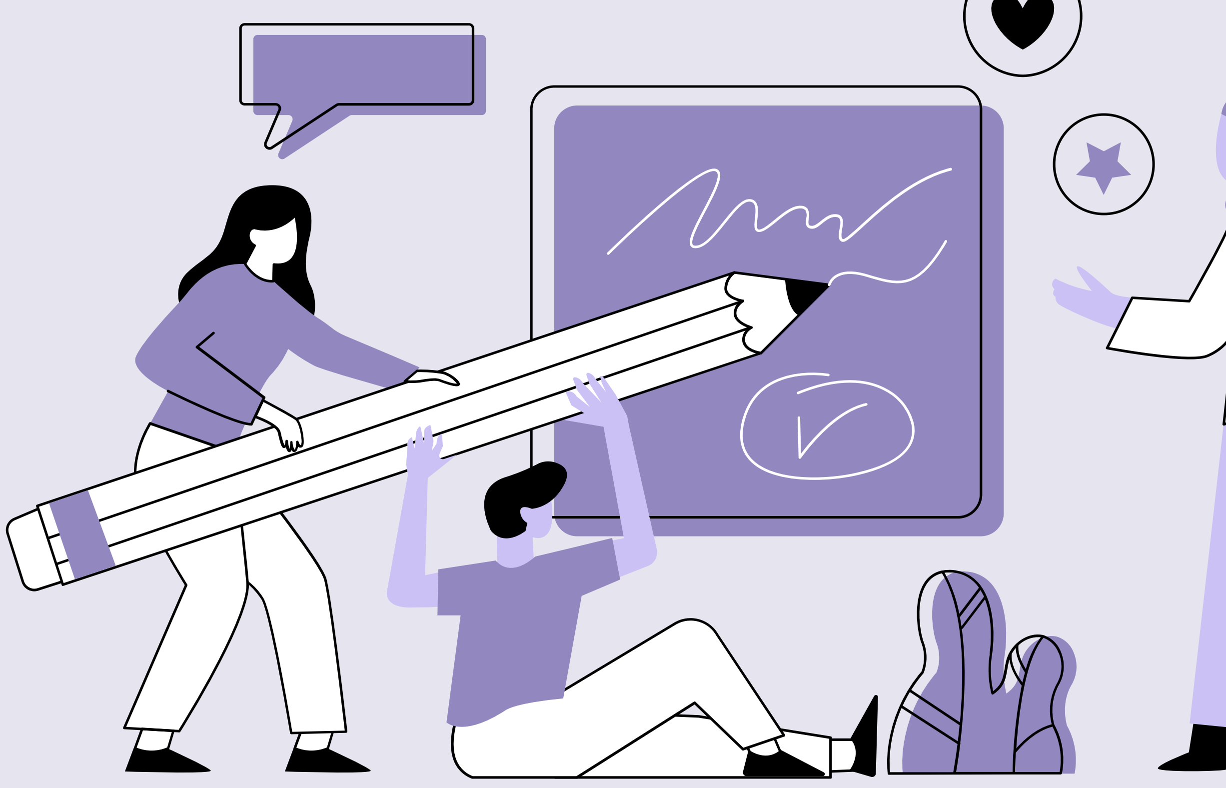

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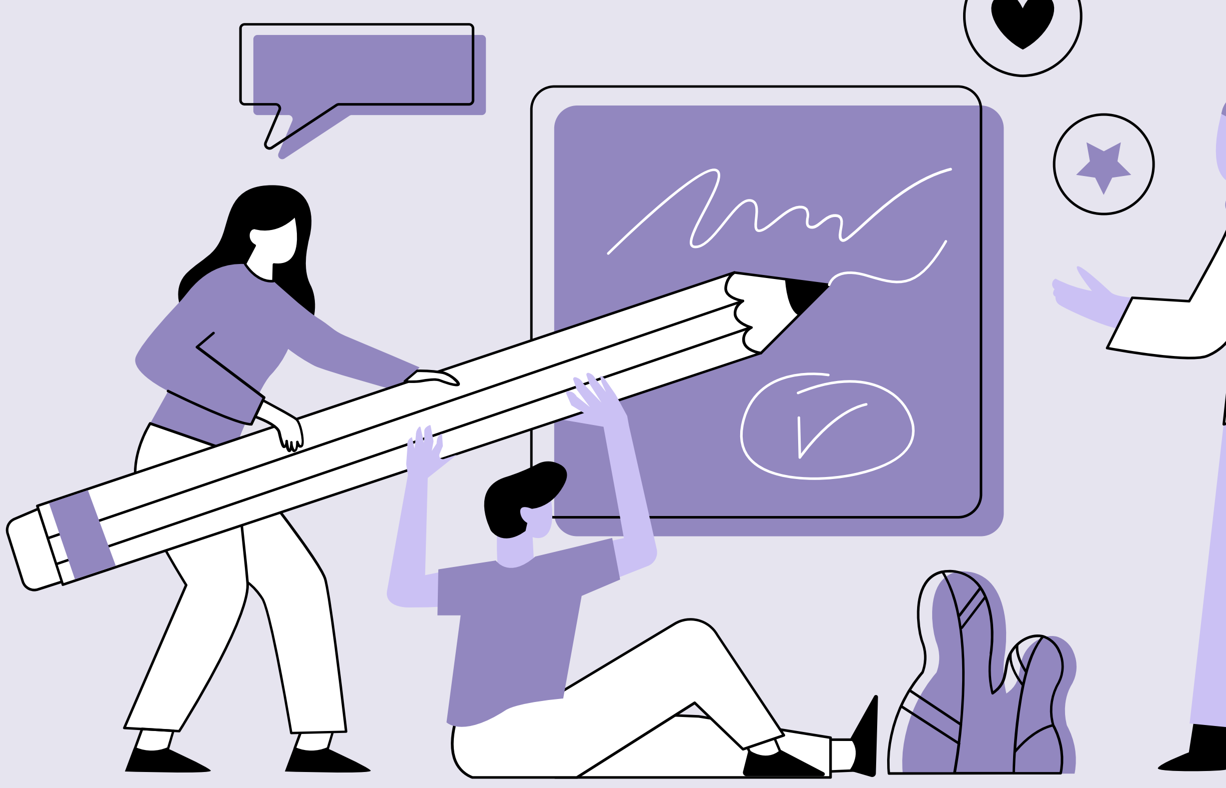


Stage |

Pre-initiation and Initiation

Project Idea

Attendance tracking in educational institutions and workplaces is currently manual, time-consuming, and prone to errors.
This business case proposes the implementation of an RFID/Fingerprint/Camera-based smart attendance system to automate and improve the accuracy, efficiency, and security of attendance tracking processes.

Business Case

1.0 Introduction/ Background

Attendance tracking in educational institutions and workplaces is currently manual, time-consuming, and prone to errors. This business case proposes the implementation of an RFID/Fingerprint/Camera-based smart attendance system to automate and improve the accuracy, efficiency, and security of attendance tracking processes.

2.0 Business Objective

The primary objective is to replace manual attendance tracking with a technology-driven solution that enhances accountability, reduces administrative effort, and improves operational efficiency.

3.0 Current Situation and Problem/Opportunity Statement

Manual attendance tracking methods are inefficient, leading to inaccuracies and payroll discrepancies.

There is an opportunity to leverage technology to automate the process, improve accuracy, and enhance accountability.

4.0 Critical Assumption and Constraints

Adequate infrastructure is available for implementing the smart attendance system. Users will adapt to the new system and accept changes in attendance tracking processes. Privacy and security concerns regarding biometric data will be addressed. The project must be implemented within the allocated budget.

5.0 Analysis of Option and Recommendation

The recommended option is to implement an integrated smart attendance system using RFID, fingerprint recognition, and cameras for accurate and secure attendance tracking

6.0 Preliminary Project Requirements

RFID readers and tags/cards for identification

Fingerprint recognition devices for biometric identification

Cameras for visual verification

Network infrastructure for real-time data transfer

Attendance management software and integration with existing systems

User training and change management initiative

7.0 Budget Estimate and Financial Analysis

A preliminary estimate of costs for the entire project is $307,555 within 7 months and 26 days detailed financial analysis including :Initiating: $15,000 ,Planning: $25,000, Executing: $225,000,Monitoring and Controlling: $30,000, Closing: $12,555 and Reserves $61,511

8.0 Schedule Estimate

The sponsor would like to see the project completed within seven months, but there is some flexibility in the schedule.

9.0 Potential Risk

Risk Category		Risk Description	Impact	Likelihood	Mitigation Strategy
Technical Challenges		Implementation difficulties, compatibility issues, and system malfunctions	High	Medium	Thorough system testing, engaging experienced technical team, and having contingency plans
Cost Overruns		Actual expenses exceeding initial budget estimates	High	Medium	Detailed cost analysis, regular budget monitoring, and setting appropriate contingency funds
Integration Issues		Complications in integrating attendance system with existing systems and workflows	Medium	Medium	Careful planning, close collaboration with IT team, and conducting thorough testing

Data Security and Privacy	Concerns around handling of biometric data and personal information	High	Medium	Implementing robust security protocols, data encryption, and strict access controls, ensuring compliance with regulations
User Adoption and Training	Employees requiring time and training to adapt to the new system	Medium	High	Comprehensive change management plan, providing extensive user training and support, and incentivizing adoption
Regulatory Compliance	Failure to comply with attendance tracking, data privacy, and security regulations	High	Medium	Continuous monitoring of relevant regulations and adapting system accordingly, seeking legal counsel
Operational Disruption	Hardware or software malfunctions leading to system downtime	High	Medium	Developing a robust backup and disaster recovery plan, ensuring regular maintenance and updates

10.0 Exhibits

Exhibit A: Financial Analysis

Financial Analysis for Project Name

Created by: **Date:**

Note: Change the inputs, shown in green below (i.e. interest rate, number of years, costs, and benefits). Be sure to double-check the formulas based on the inputs.

Discount rate	8.00%					
Assume the project is completed in Year 0			**Year**			
	0	1	2	3	**Total**	
Costs	200,000	50,000	29,555	28,000		
Discount factor	1.00	0.93	0.86	0.79		
Discounted costs	**200,000**	**46,500**	**25,417**	**22,120**	**294,037**	
Benefits	0	200,000	200,000	200,000		
Discount factor	1.00	0.93	0.86	0.79		
Discounted benefits	**0**	**186,000**	**172,000**	**158,000**	**516,000**	
Discounted benefits - costs	(200,000)	139,500	146,583	135,880	**221,963**	**NPV**
Cumulative benefits - costs	(200,000)	(60,500)	86,083	221,963		
ROI	**75%**					
	Payback in Year 1					

Assumptions

Enter assumptions here

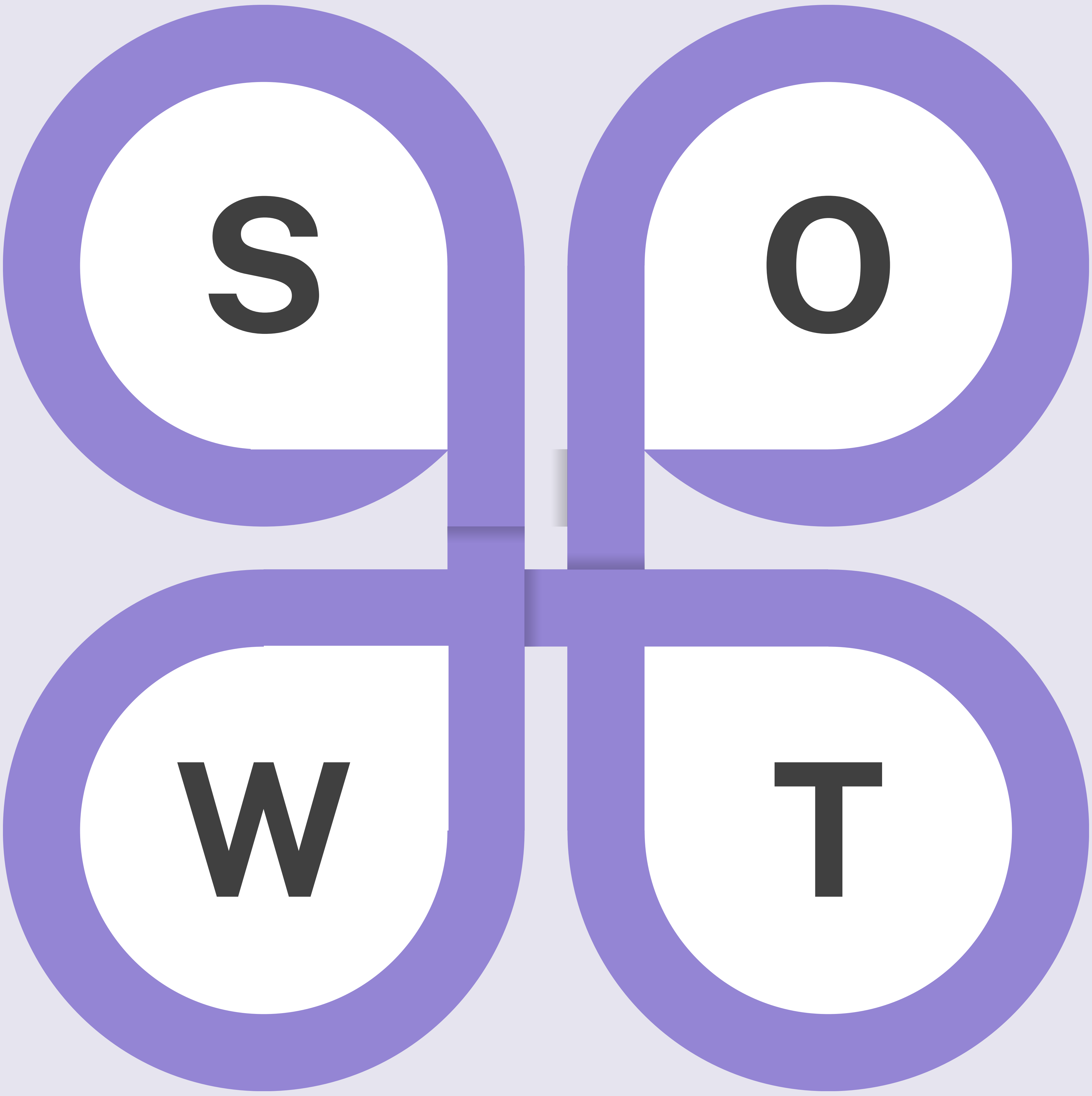
SWOT

1. Strengths:

- Automation and Efficiency
- Enhanced Data Accuracy
- Improved Accountability
- Potential for Cost Savings
- Integration Potential with Existing Systems

2. Opportunities

- Streamlined Attendance Tracking Processes
- Improved Resource Allocation
- Enhanced Data Analysis and Reporting Capabilities
- Scalability for Future Needs



3. Weaknesses

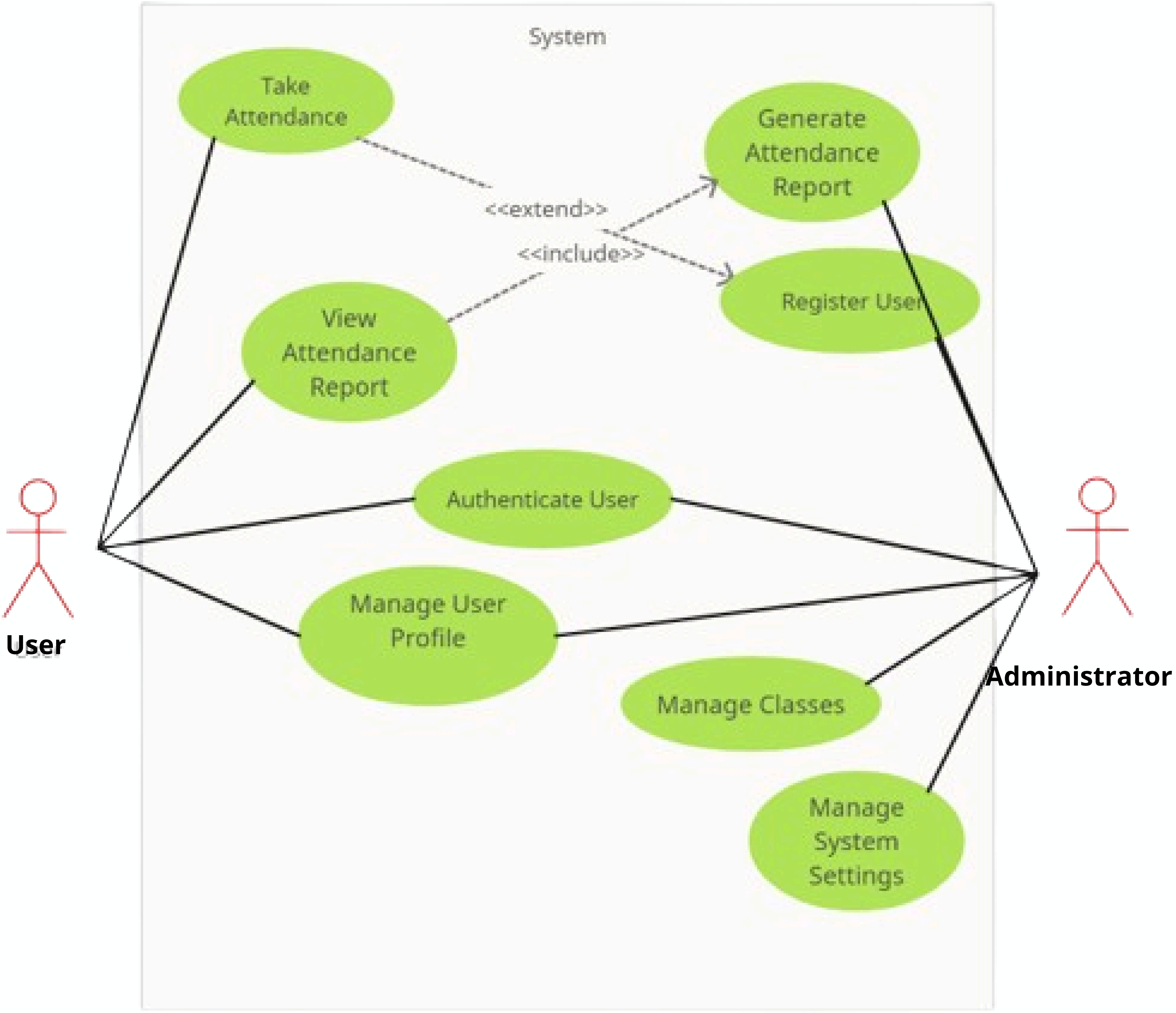
- Technical Challenges and Implementation Risks
- Resistance to Change
- Initial Investment Cost
- Privacy and Security Concerns

4. Threats

- Potential Disruption during Implementation
- Data Breaches and Security Risks
- Lack of User Adoption and Training
- Regulatory Compliance and Legal Issues

software requirement specification

Use Case Diagram



Use-Case ID:	UC1		
Use-Case Name:	Register User		
Created By:	Administrator	**Last Updated By:**	Administrator
Date Created:	2024-05-05	**Date Last Updated:**	2024-05-05

Actors:	Administrator
Description:	Allows the Administrator to register a new user in the system.
Preconditions:	The Administrator must be authenticated.
Postconditions:	The user is registered in the system.
Normal Flow:	1. Administrator selects the "Register User" option in the system. 2. System presents a registration form. 3. Administrator fills in the required user details (e.g., name, email, password). 4. Administrator submits the registration form. 5. System validates the entered information. 6. System creates a new user account with the provided details. 7. System confirms the successful registration and displays a success message.
Alternative Flow:	6a. If the entered information is invalid or incomplete, the system displays error messages and prompts the Administrator to correct the form.
Exceptions Flow:	none
Includes:	none
Special Requirements:	• The registration form should include mandatory fields and validation rules. • The system should enforce secure password requirements.
Assumptions:	• The Administrator has the necessary privileges to register new users. • The system has a database in place to store user information.
Notes and Issues:	none

Use-Case ID:	UC2		
Use-Case Name:	Authenticate User		
Created By:	Administrator	**Last Updated By:**	Administrator
Date Created:	2024-05-05	**Date Last Updated:**	2024-05-05

Actors:	Administrator, User
Description:	Authenticates the user's credentials to log in to the system.
Preconditions:	none
Postconditions:	The user is logged in to the system.
Normal Flow:	1. User opens the login page of the system. 2. User enters their username and password. 3. User submits the login form. 4. System verifies the entered credentials. 5. If the credentials are valid, the system logs in the user and displays the home page. 6. If the credentials are invalid, the system displays an error message and prompts the user to reenter the login details..
Alternative Flow:	none
Exceptions Flow:	none
Includes:	none
Special Requirements:	• The system should securely store and verify user credentials. • The system should implement appropriate measures to prevent unauthorized access
Assumptions:	• Users have been previously registered in the system.
Notes and Issues:	none

Use-Case ID:	UC3		
Use-Case Name:	Take Attendance		
Created By:	Administrator	**Last Updated By:**	Administrator
Date Created:	2024-05-05	**Date Last Updated:**	2024-05-05

Actors:	Administrator
Description:	Allows the Administrator to take attendance for a class.
Preconditions:	The Administrator must be authenticated and have access to the class.
Postconditions:	Attendance for the class is recorded..
Normal Flow:	1. Administrator selects the "Take Attendance" option for a specific class in the system. 2. System displays the list of enrolled students for the selected class. 3. Administrator marks the attendance for each student (e.g., present, absent) using appropriate controls. 4. Administrator submits the attendance. 5. System records the attendance for the class
Alternative Flow:	none
Exceptions Flow:	none
Includes:	UC1 (Register User)
Special Requirements:	• The system should provide an intuitive interface for marking attendance. • The system should store and maintain attendance records accurately.
Assumptions:	• Class and student details have been previously added to the system.
Notes and Issues:	none

Use-Case ID:	UC4		
Use-Case Name:	Manage User Profile		
Created By:	Administrator	**Last Updated By:**	Administrator
Date Created:	2024-05-05	**Date Last Updated:**	2024-05-05

Actors:	Administrator
Description:	Allows the Administrator to manage their own user profile.
Preconditions:	The Administrator must be authenticated.
Postconditions:	The Administrator's profile is updated.
Normal Flow:	1. Administrator selects the "Manage User Profile" option in the system. 2. System displays the Administrator's profile information. 3. Administrator modifies the desired fields (e.g., name, email, password). 4. Administrator saves the changes. 5. System updates the Administrator's profile with the modified information.
Alternative Flow:	none
Exceptions Flow:	none
Includes:	none
Special Requirements:	• The system should provide appropriate validation and security measures for profile updates. • The system should allow the Administrator to change their password securely.
Assumptions:	• The Administrator has the necessary privileges to manage their own profile.
Notes and Issues:	none

Use-Case ID:	UC5		
Use-Case Name:	Manage Classes		
Created By:	Administrator	**Last Updated By:**	Administrator
Date Created:	2024-05-05	**Date Last Updated:**	2024-05-05

Actors:	Administrator
Description:	Allows the Administrator to manage classes (create, update, and delete).
Preconditions:	The Administrator must be authenticated.
Postconditions:	The class information is created, updated, or deleted in the system.
Normal Flow:	1. Administrator selects the "Manage Classes" option in the system. 2. System displays the list of existing classes. 3. Administrator chooses to create a new class or selects an existing class to update or delete. 4. If creating a new class: a. Administrator enters the details for the new class (e.g., class name, time, location). b. Administrator saves the new class. c. System adds the new class to the list of classes. 5. If updating an existing class: a. Administrator modifies the desired fields of the selected class. b. Administrator saves the changes. c. System updates the class information. 6. If deleting an existing class: a. Administrator confirms the deletion. b. System removes the class from the list of classes.
Alternative Flow:	none
Exceptions Flow:	none
Includes:	none
Special Requirements:	• The system should enforce any constraints or business rules related to class management. • The system should provide appropriate notifications or warnings for potential conflicts or errors.
Assumptions:	• The system should enforce any constraints or business rules related to class management. • The system should provide appropriate notifications or warnings for potential conflicts or errors.
Notes and Issues:	none

Use-Case ID:	UC6		
Use-Case Name:	Generate Attendance Report		
Created By:	Administrator	**Last Updated By:**	Administrator
Date Created:	2024-05-05	**Date Last Updated:**	2024-05-05

Actors:	Administrator
Description:	Allows the Administrator to generate an attendance report for a specific class.
Preconditions:	The Administrator must be authenticated and have access to the class.
Postconditions:	An attendance report is generated and made available.
Normal Flow:	1. Administrator selects the "Generate Attendance Report" option for a specific class in the system. 2. System presents options to specify the date range or other report parameters. 3. Administrator selects the desired parameters for the report. 4. Administrator initiates the report generation. 5. System processes the request and generates the attendance report. 6. System presents the generated attendance report to the Administrator for viewing or download.
Alternative Flow:	none
Exceptions Flow:	none
Includes:	View Attendance Report
Special Requirements:	• The system should provide various report customization options • The system should generate accurate and informative attendance reports..
Assumptions:	• Attendance data for the specified class and time period is available in the system.
Notes and Issues:	none

Functional Requirements

User Registration	The system shall allow employees to register their information, including personal details and identification data (such as fingerprints or RFID cards).
Authentication	The system shall authenticate the identity of employees using RFID, fingerprint recognition, or camera-based verification methods.
Attendance Recording	The system shall record attendance for employees using RFID, fingerprint recognition, or camera-based technology, capturing accurate timestamps.
Real-Time Monitoring	The system shall provide real-time monitoring of attendance records, allowing authorized personnel to view attendance data as it is recorded.
Attendance Reporting	The system shall generate attendance reports for individual employees, as well as overall attendance statistics for specific time periods.
Integration with Existing Systems	The system shall have the capability to integrate with existing employee management systems to ensure accurate record-keeping and data synchronization.
Access Control	The system shall control access to restricted areas based on the recorded attendance data, granting or denying entry as per authorization rules.
Attendance History	The system shall store historical attendance data for future reference and analysis, allowing administrators to track attendance patterns over time

Non-functional Requirements

Security	The system shall employ strong encryption techniques to protect sensitive attendance data and prevent unauthorized access.
Accuracy	The system shall have a high level of accuracy in capturing attendance data to ensure reliable attendance records.
Speed and Efficiency	The system shall record attendance quickly and efficiently, minimizing delays and providing a seamless user experience.
Reliability	The system shall have a high level of reliability, ensuring that attendance records are consistently and accurately captured without system failures.
Data Backup and Recovery	The system shall regularly backup attendance data to prevent data loss and have a reliable data recovery mechanism in case of system failures.

Stakeholder Register

Name	Position	Internal/ External	Project Role	Contact Information
Wed Abu Zinadah	Project Manager	Internal	Project Manager	+966555898754, wtabozenada@uj.edu.sa
Rasha Alomari	Managing organization	Internal	Human Resources	rmalamri@uj.edu.sa
Mohammed	Internal Resource Management	Internal	Resource Management	Mohammed.uj.ed.sa
|Prof. Abdulaziz	Advisor	Internal	Computer Science Department Advisor	prof.abdulaziz@ju.edu.sa
Raghad Al-saiari	Software Developer	Internal	Team Member	2210207@uj.edu.sa
Deema Alsini	Quality Assurance Engineer	Internal	Team Member	2215727@uj.edu.sa
Sarah Bahashwan	System Analyst	Internal	Team Member	2211101@uj.edu.sa
Rabia Hilal	Security Specialist	Internal	Team Member	2210864@uj.edu.sa
Ahmed	Testing Specialist	External	Testing Team	ahmed.test@gmail.com
Mr. Khalid	Teacher Association	Internal	Stakeholder	khalid.Teacher@uj.edu.sa

Stakeholder Strategy

Name	Level of Interest	Level of Influence	Potential Management Strategies
Wed Abu Zinadah	High	High	Frequent engagement through meetings, updates, and feedback Involve in key decisions and strategic planning Align project objectives with stakeholder's interests
Rasha Alomari	High	High	Regular communication through meetings, email, and status reports Collaborate on resource allocation, budgeting, and HR management Involve in risk identification and mitigation
Mohammed	Medium	Medium	Provide periodic updates through email, meetings, and reports Solicit input on resource decisions Keep informed of progress and changes
Prof. Abdulaziz	High	High	Frequent one-on-one meetings to discuss direction, risks, and challenges Seek expert advice on procurement, legal, and regulatory matters Involve in project governance and control development
Raghad Al-saiari	High	Medium	Maintain regular communication through meetings, updates, and discussions Involve in design, development, and testing of deliverables Solicit feedback on user experience and requirements
Deema Alsini	High	Medium	Engage regularly through meetings, QA reviews, and discussions Collaborate on test plans, test cases, and quality control Seek input on technical issues and risks
Sarah Bahashwan	High	Medium	Provide frequent updates on progress, challenges, and changes Involve in requirements analysis, system design, and implementation Leverage her expertise in system analysis
Rabia hilal	High	Medium	Maintain regular communication on security concerns, risks, and mitigation Involve in the development of security policies and incident response Collaborate on the implementation and testing of security measures
Mr. Khalid	Medium	Medium	Provide periodic updates on project progress and impact on teaching Solicit feedback on the project's impact on teaching activities Collaborate on training programs and resources for faculty and staff
Ahmed	Medium	Low	Provide regular updates on testing activities, issues, and delays Involve in the development and execution of test plans and procedures Recognize and reward his contributions to quality assurance

Project Charter

Project Charter

Project Title:
RFID/Fingerprint/Camera based smart student and employee attendance system

Project Start Date: 17-5-2024 **Projected Finish Date:** 21-3-2025

Budget Information :

The project has been allocated a budget of 307,555$. The majority of the costs will be associated with the procurement of hardware, software licenses, and internal labor costs.

Project Manager:

Wed Abu Zinadah , +966555898754, wtabozenada@uj.edu.sa

Project Objectives:

- To design and implement a comprehensive attendance system using RFID, Fingerprint, and Camera technologies.
- To enhance the accuracy and efficiency of attendance tracking for students and employees.
- To provide real-time attendance updates to relevant stakeholders.

Main Project Success Criteria:

- Successful implementation and integration of RFID, Fingerprint, and Camera technologies into a unified attendance system.
- Achieving an attendance tracking accuracy rate of 95% or higher.
- Receiving positive feedback from users (students and employees) and stakeholders (school administration and parents).

Approach:

- Conduct a needs assessment to understand the specific requirements of the users and stakeholders.
- Procure necessary hardware and software for RFID, Fingerprint, and Camera technologies.
- Design and develop the attendance system, ensuring seamless integration of all technologies.
- Test the system thoroughly to ensure accuracy and efficiency.
- Deploy the system and provide necessary training to users and stakeholders.

Roles and Responsibilities :

Role	Name	Organization/ Position	Contact Information
Project Manager	Wed Abu Zinadah	Project Manager	wtabozenada@uj.edu.sa
Human Resources	Rasha Alomari	Managing organization	rmalamri@uj.edu.sa
Resource Manager	Mohammed	Internal Resource Management	Mohammed.uj.ed.sa
Advisor	Prof. Abdulaziz	Procurement Specialist	prof.abdulaziz@ju.edu.sa
Project Team	Raghad Al-saiari	Software Developer	2210207@uj.edu.sa
Development Team	Deema Alsini	Quality Assurance Engineers	2215727 @uj.edu.sa
Training Specialist	Sarah Bahashwan	System Analyst	2211101@uj.edu.sa
Change Management Specialist	Rabia hilal	Security Specialist	2210864@uj.edu.sa
Finance Department	Jeddah university	Funding Organization	Mail.uj.edu.sa
Stakeholder	Mr.Khalid	Teacher Association	khalid.Teacher @uj.edu.sa
Testing Team	Ahmed	Testing Specialist	ahemd.test@gmail.com

Sign-off: (Signatures of all above stakeholders. Can sign by their names in table above.)

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

Wed Abu Zinadah: Looking forward to the successful completion of this project.

Mr.Khalid: Teachers are excited about this new system.

Prof.Abdulaziz: Hope this system will bring transparency in attendance tracking.

Kick-off Meeting

Kick-off Meeting

Meeting Objective: Get the project off to a great start by introducing key stakeholders, reviewing project goals, and discussing future plans

Agenda:

- Introductions of attendees
- Background of project
- Review of project-related documents (i.e. business case, project charter)
- Discussion of project organizational structure
- Discussion of project scope, time, and cost goals
- Discussion of other important topics
- List of action items from meeting

Action Item	Assigned To	Due Date
Review and finalize project charter	Wed Abu Zinadah	May 25, 2024
Software Development	Raghad Al-saiari	May 27, 2024
Quality Assurance	Deema Alsini	May 25, 2024
System Analysis	Sarah Bahashwan	May 25, 2024
Create project schedule and timeline	Wed Abu Zinadah	May 26, 2024
Define and document project scope	Wed Abu Zinadah	May 27, 2024
Security Assessment	Rabia Hilal	May 28, 2024
Establish regular project status meetings	Wed Abu Zinadah	May 25, 2024

Date and time of next meeting:
25/Jun/2024

Stage 2

Planning Stage of Scope, Time, and Cost

Scope Statement

Project Justification:
The reason why we need this project is aims to automate and improve the accuracy, efficiency, and security of attendance tracking processes in educational institutions and workplaces

Product Characteristics and Requirements:
1. The product should be able to accurately track attendance using RFID, Fingerprint, and Camera technologies.
2. The product should be user-friendly and easy to use for both students and employees.
3. The product should be able to provide real-time attendance updates to relevant stakeholders.
4. The product should ensure data security and privacy compliance.

Summary of Project Deliverables Project management-related deliverables: business case, charter, team contract, scope statement, WBS, schedule, cost baseline, status reports, final project presentation, final project report, lessons-learned report, and any other documents required to manage the project.
Product-related deliverables: research reports, design documents, software code, hardware, etc.
1. RFID readers and tags/cards for identification: The system will include RFID readers and tags/cards for identification when the system is first implemented. The project team will decide on the initial quantity based on the needs assessment.
2. Fingerprint recognition devices for biometric identification: The system will include fingerprint recognition devices for biometric identification. Real-world examples of successful implementations will be provided for reference.
3. Cameras for visual verification: The system will include cameras for visual verification. Instructions for their use and maintenance will be provided, along with examples of effective camera placement and usage.
4. Network infrastructure for real-time data transfer: The system will require a robust network infrastructure for real-time data transfer. The project team will ensure compatibility with existing network systems.
5. Attendance management software and integration with existing systems: The system will include attendance management software that will be integrated with existing systems. User training and support materials will be provided to facilitate a smooth transition.
6. User training and change management initiatives: The project will include user training and change management initiatives to ensure successful adoption of the new system. Feedback sessions will be conducted to monitor satisfaction levels and make necessary adjustments.
7. Ongoing Maintenance and Technical Support: The project will include provisions for ongoing maintenance and technical support to address any potential issues and ensure uninterrupted service.

Project Success Criteria:
- Successful implementation and integration of RFID, Fingerprint, and Camera technologies into a unified attendance system.
- Achieving an attendance tracking accuracy rate of 95% or higher.
- Receiving positive feedback from users (students and employees) and stakeholders (school administration and parents).

Work Breakdown Structure

Work Breakdown Structure

1.0 Initiating
- 1.1 Define project team members
- 1.2 Determine which companies to cooperate with for hardware and software
- 1.3 Prepare project charter
- 1.4 Hold project kick-off meeting

2.0 Planning
- 2.1 Hold team planning meeting
  - 2.1.1 Determine the responsibilities of each team member
- 2.2 Prepare scope statement
- 2.3 Prepare WBS
- 2.4 Prepare schedule and cost baseline
  - 2.4.1 Preparing the project start and end schedule
  - 2.4.2 Explain the steps of the project
  - 2.4.3 Determine the duration of each step
  - 2.4.4 Create draft Gantt chart
  - 2.4.5 Clarify the programs used and their cost

3.0 Executing
- 3.1 Procurement
  - 3.1.1 Procure RFID readers and tags/cards
  - 3.1.2 Procure Fingerprint recognition devices
  - 3.1.3 Procure Cameras
- 3.2 Development
  - 3.2.1 Develop Attendance management software
  - 3.2.2 Integrate with existing systems
- 3.3 Training
  - 3.3.1 Conduct User Training
  - 3.3.2 Implement Change Management Initiatives
- 3.4 Testing
  - 3.4.1 Conduct System Testing

4.0 Monitoring and Controlling
- 4.1 Resource Management
- 4.2 Cost Management
- 4.3 Progress reports
- 4.4 Change requests

5.0 Closing
- 5.1 Prepare final project report
- 5.2 Prepare final project presentation
- 5.3 Lessons learned

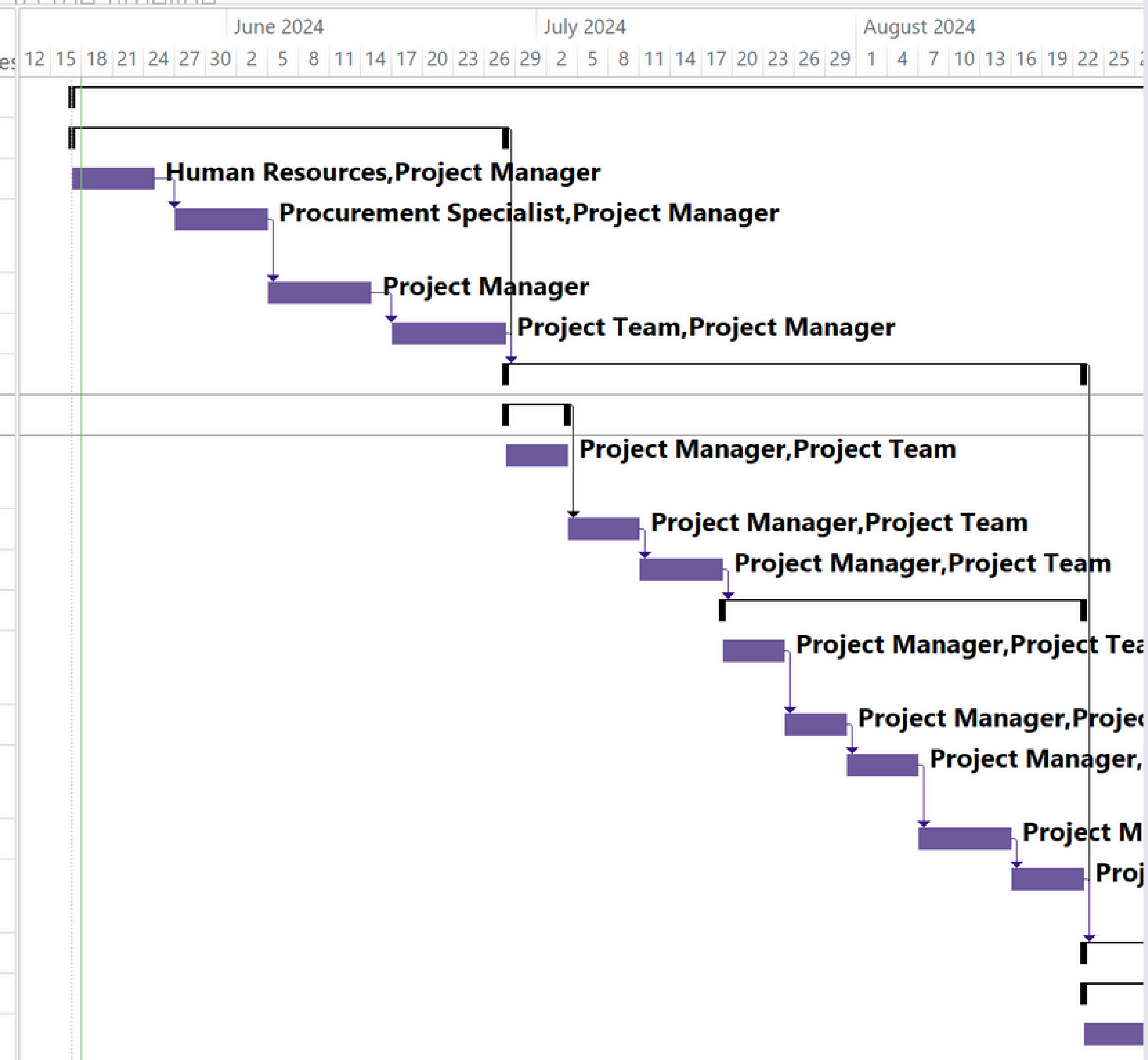
Grantt Chart

Sat 9/14/24

Jun '24 | Jul '24 | Aug '24 | Sep '24 | Oct '24 | Nov '24 | Dec '24 | Jan '25 | Feb '25 | Mar '25

Add tasks with dates to the timeline

Task Name	Duration	Start	Finish	Predecessors
project	**221 days**	**Fri 5/17/24**	**Fri 3/21/25**	
1.0 Initiating	**30 days**	**Fri 5/17/24**	**Thu 6/27/24**	
1.1 Define project team members	6 days	Fri 5/17/24	Fri 5/24/24	
1.2 Determine which companies to cooperate with for hardware and software	7 days	Mon 5/27/24	Tue 6/4/24	3
1.3 Prepare project charter	8 days	Wed 6/5/24	Fri 6/14/24	4
1.4 Hold project kick-off meeting	9 days	Mon 6/17/24	Thu 6/27/24	5
2.0 Planning	**40 days**	**Fri 6/28/24**	**Thu 8/22/24**	**2,6**
2.1 Hold team planning meeting	**4 days**	**Fri 6/28/24**	**Wed 7/3/2**	
2.1.1 Determine the responsibilities of each team member	4 days	Fri 6/28/24	Wed 7/3/24	
2.2 Prepare scope statement	5 days	Thu 7/4/24	Wed 7/10/24	8
2.3 Prepare WBS	6 days	Thu 7/11/24	Thu 7/18/24	10
2.4 Prepare schedule and cost baseline	**25 days**	**Fri 7/19/24**	**Thu 8/22/24**	**11**
2.4.1 Preparing the project start and end schedule	4 days	Fri 7/19/24	Wed 7/24/24	
2.4.2 Explain the steps of the project	4 days	Thu 7/25/24	Tue 7/30/24	13
2.4.3 Determine the duration of each step	5 days	Wed 7/31/24	Tue 8/6/24	14
2.4.4 Create draft Gantt chart	7 days	Wed 8/7/24	Thu 8/15/24	15
2.4.5 Clarify the programs used and their cost	5 days	Fri 8/16/24	Thu 8/22/24	16
3.0 Executing	**90 days**	**Fri 8/23/24**	**Thu 12/26/24**	**7,17**
3.1 Procurement	**20 days**	**Fri 8/23/24**	**Thu 9/19/24**	
3.1.1 Procure RFID readers and tags/cards	7 days	Fri 8/23/24	Mon 9/2/24	

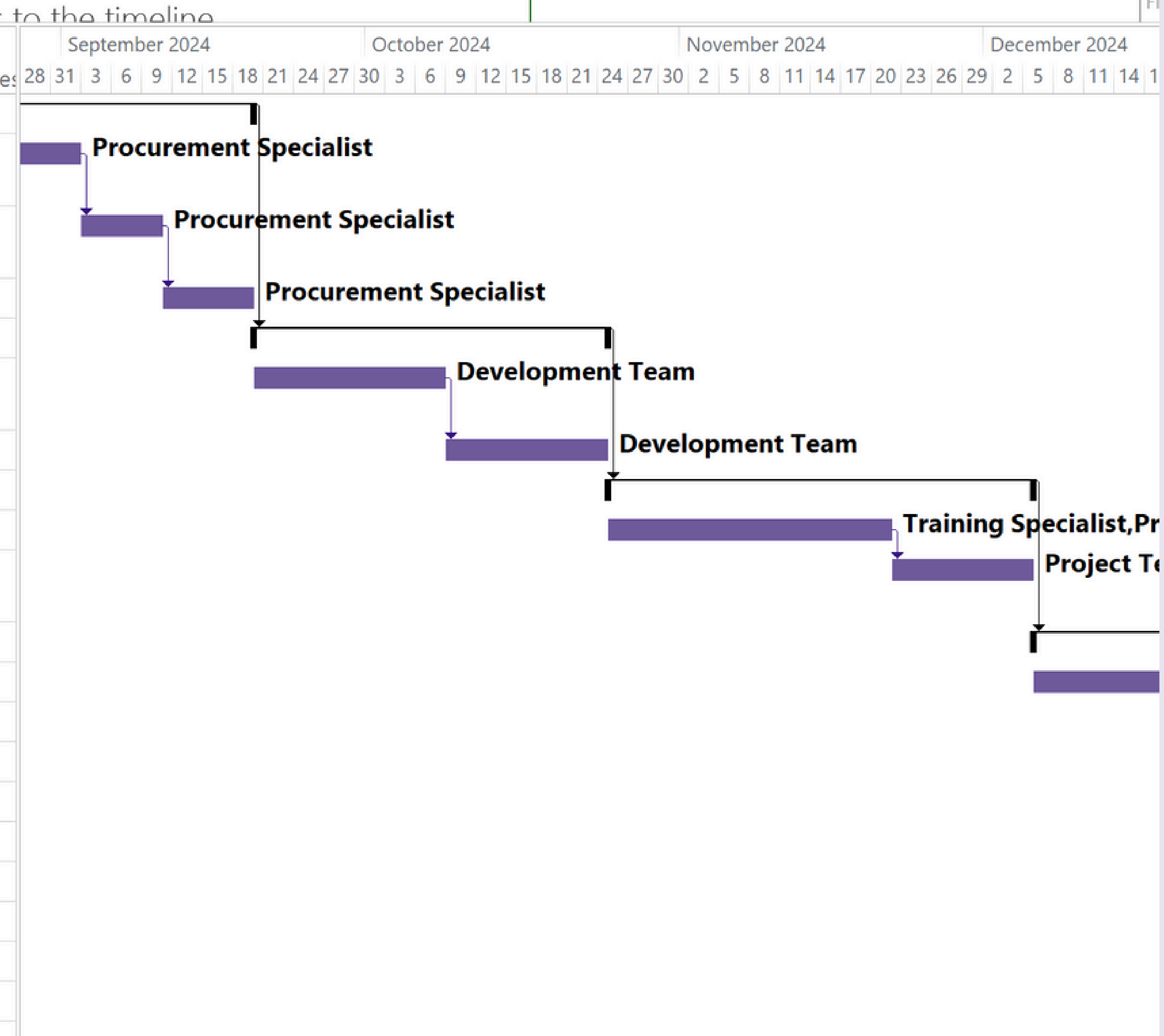


Wed 8/28/24 | Tue 12/31/24

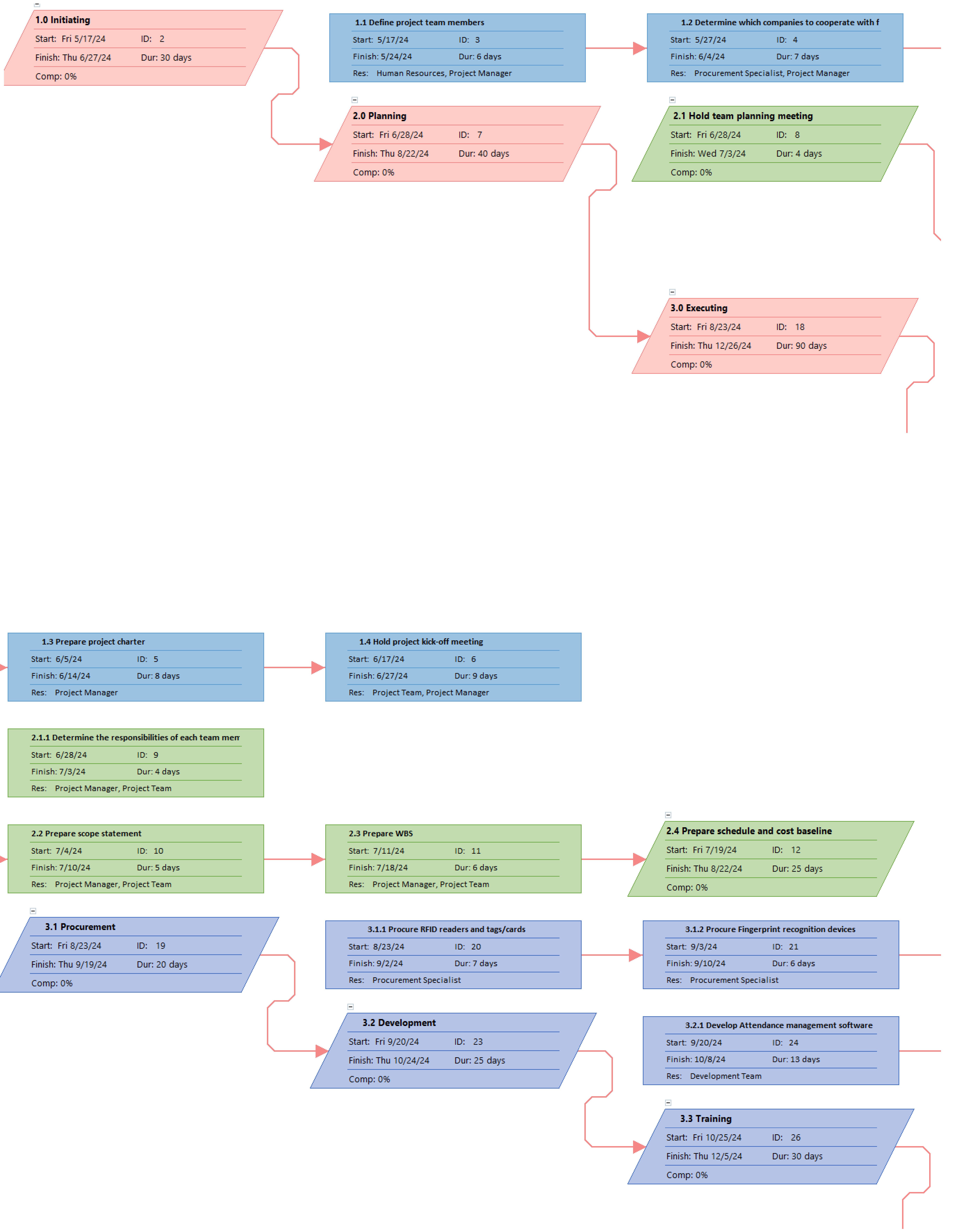
Jun '24 | Jul '24 | Aug '24 | Sep '24 | Oct '24 | Nov '24 | Dec '24 | Jan '25 | Feb '25 | Mar '25

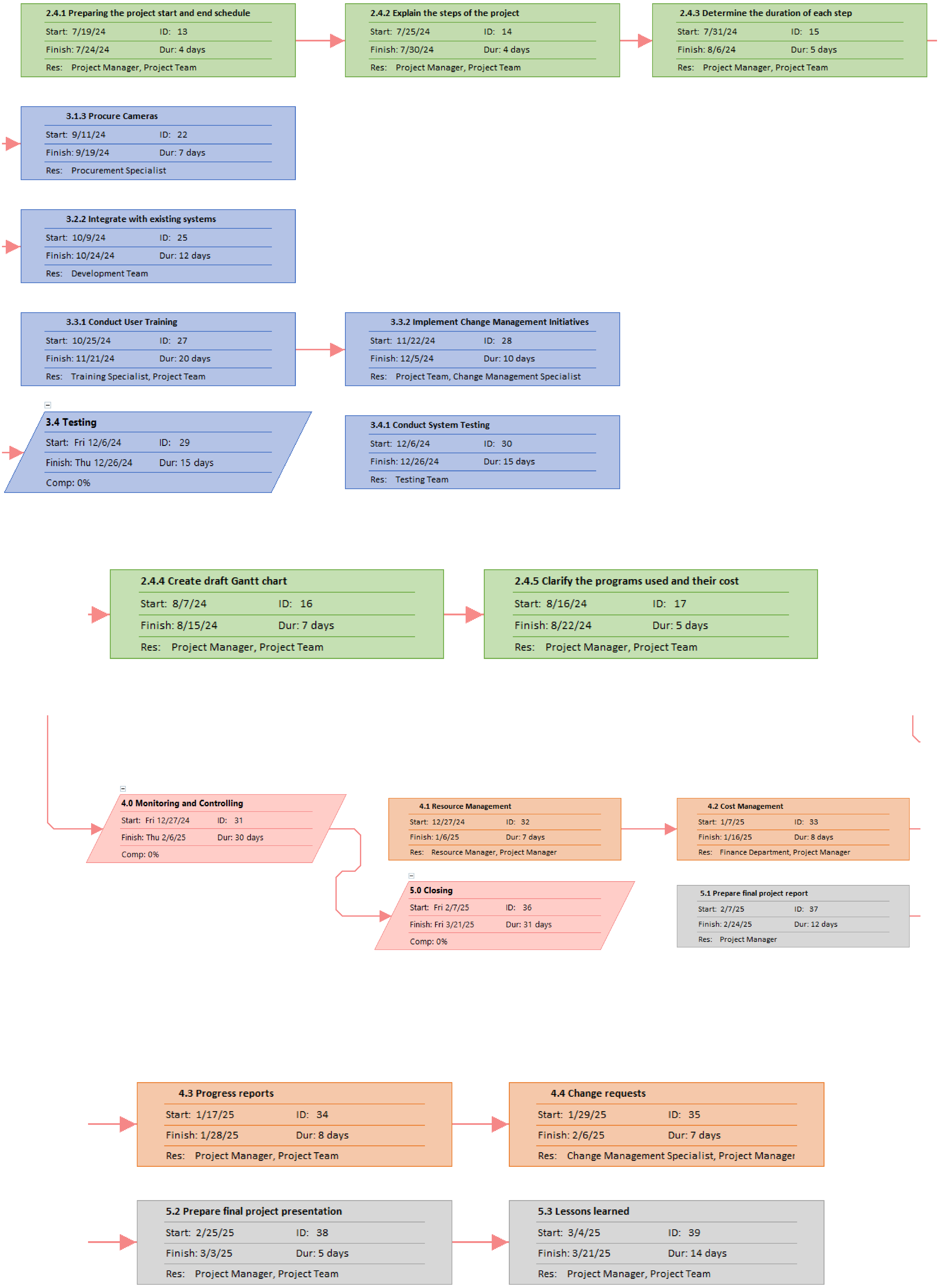
Add tasks with dates to the timeline

Task Name	Duration	Start	Finish	Predecessors
3.1 Procurement	**20 days**	**Fri 8/23/24**	**Thu 9/19/24**	
3.1.1 Procure RFID readers and tags/cards	7 days	Fri 8/23/24	Mon 9/2/24	
3.1.2 Procure Fingerprint recognition devices	6 days	Tue 9/3/24	Tue 9/10/24	20
3.1.3 Procure Cameras	7 days	Wed 9/11/24	Thu 9/19/24	21
3.2 Development	**25 days**	**Fri 9/20/24**	**Thu 10/24/24**	**19**
3.2.1 Develop Attendance management software	13 days	Fri 9/20/24	Tue 10/8/24	
3.2.2 Integrate with existing systems	12 days	Wed 10/9/24	Thu 10/24/24	24
3.3 Training	**30 days**	**Fri 10/25/24**	**Thu 12/5/24**	**23**
3.3.1 Conduct User Training	20 days	Fri 10/25/24	Thu 11/21/24	
3.3.2 Implement Change Management Initiatives	10 days	Fri 11/22/24	Thu 12/5/24	27
3.4 Testing	**15 days**	**Fri 12/6/24**	**Thu 12/26/24**	**26**
3.4.1 Conduct System Testing	15 days	Fri 12/6/24	Thu 12/26/24	
4.0 Monitoring and Controlling	**30 days**	**Fri 12/27/24**	**Thu 2/6/25**	**18,30**
4.1 Resource Management	7 days	Fri 12/27/24	Mon 1/6/25	
4.2 Cost Management	8 days	Tue 1/7/25	Thu 1/16/25	32
4.3 Progress reports	8 days	Fri 1/17/25	Tue 1/28/25	33
4.4 Change requests	7 days	Wed 1/29/25	Thu 2/6/25	34
5.0 Closing	**31 days**	**Fri 2/7/25**	**Fri 3/21/25**	**31,35**
5.1 Prepare final project report	12 days	Fri 2/7/25	Mon 2/24/25	
5.2 Prepare final project presentation	5 days	Tue 2/25/25	Mon 3/3/25	37
5.3 Lessons learned	14 days	Tue 3/4/25	Fri 3/21/25	38



Network Diagram


1.0 Initiating
Start: Fri 5/17/24 ID: 2
Finish: Thu 6/27/24 Dur: 30 days
Comp: 0%
1.1 Define project team members
Start: 5/17/24 ID: 3
Finish: 5/24/24 Dur: 6 days
Res: Human Resources, Project Manager
1.2 Determine which companies to cooperate with f
Start: 5/27/24 ID: 4
Finish: 6/4/24 Dur: 7 days
Res: Procurement Specialist, Project Manager
2.0 Planning
Start: Fri 6/28/24 ID: 7
Finish: Thu 8/22/24 Dur: 40 days
Comp: 0%
2.1 Hold team planning meeting
Start: Fri 6/28/24 ID: 8
Finish: Wed 7/3/24 Dur: 4 days
Comp: 0%
3.0 Executing
Start: Fri 8/23/24 ID: 18
Finish: Thu 12/26/24 Dur: 90 days
Comp: 0%
1.3 Prepare project charter
Start: 6/5/24 ID: 5
Finish: 6/14/24 Dur: 8 days
Res: Project Manager
1.4 Hold project kick-off meeting
Start: 6/17/24 ID: 6
Finish: 6/27/24 Dur: 9 days
Res: Project Team, Project Manager
2.1.1 Determine the responsibilities of each team mem
Start: 6/28/24 ID: 9
Finish: 7/3/24 Dur: 4 days
Res: Project Manager, Project Team
2.2 Prepare scope statement
Start: 7/4/24 ID: 10
Finish: 7/10/24 Dur: 5 days
Res: Project Manager, Project Team
2.3 Prepare WBS
Start: 7/11/24 ID: 11
Finish: 7/18/24 Dur: 6 days
Res: Project Manager, Project Team
2.4 Prepare schedule and cost baseline
Start: Fri 7/19/24 ID: 12
Finish: Thu 8/22/24 Dur: 25 days
Comp: 0%
3.1 Procurement
Start: Fri 8/23/24 ID: 19
Finish: Thu 9/19/24 Dur: 20 days
Comp: 0%
3.1.1 Procure RFID readers and tags/cards
Start: 8/23/24 ID: 20
Finish: 9/2/24 Dur: 7 days
Res: Procurement Specialist
3.1.2 Procure Fingerprint recognition devices
Start: 9/3/24 ID: 21
Finish: 9/10/24 Dur: 6 days
Res: Procurement Specialist
3.2 Development
Start: Fri 9/20/24 ID: 23
Finish: Thu 10/24/24 Dur: 25 days
Comp: 0%
3.2.1 Develop Attendance management software
Start: 9/20/24 ID: 24
Finish: 10/8/24 Dur: 13 days
Res: Development Team
3.3 Training
Start: Fri 10/25/24 ID: 26
Finish: Thu 12/5/24 Dur: 30 days
Comp: 0%


2.4.1 Preparing the project start and end schedule
Start: 7/19/24 ID: 13
Finish: 7/24/24 Dur: 4 days
Res: Project Manager, Project Team
2.4.2 Explain the steps of the project
Start: 7/25/24 ID: 14
Finish: 7/30/24 Dur: 4 days
Res: Project Manager, Project Team
2.4.3 Determine the duration of each step
Start: 7/31/24 ID: 15
Finish: 8/6/24 Dur: 5 days
Res: Project Manager, Project Team
3.1.3 Procure Cameras
Start: 9/11/24 ID: 22
Finish: 9/19/24 Dur: 7 days
Res: Procurement Specialist
3.2.2 Integrate with existing systems
Start: 10/9/24 ID: 25
Finish: 10/24/24 Dur: 12 days
Res: Development Team
3.3.1 Conduct User Training
Start: 10/25/24 ID: 27
Finish: 11/21/24 Dur: 20 days
Res: Training Specialist, Project Team
3.3.2 Implement Change Management Initiatives
Start: 11/22/24 ID: 28
Finish: 12/5/24 Dur: 10 days
Res: Project Team, Change Management Specialist
3.4 Testing
Start: Fri 12/6/24 ID: 29
Finish: Thu 12/26/24 Dur: 15 days
Comp: 0%
3.4.1 Conduct System Testing
Start: 12/6/24 ID: 30
Finish: 12/26/24 Dur: 15 days
Res: Testing Team
2.4.4 Create draft Gantt chart
Start: 8/7/24 ID: 16
Finish: 8/15/24 Dur: 7 days
Res: Project Manager, Project Team
2.4.5 Clarify the programs used and their cost
Start: 8/16/24 ID: 17
Finish: 8/22/24 Dur: 5 days
Res: Project Manager, Project Team
4.0 Monitoring and Controlling
Start: Fri 12/27/24 ID: 31
Finish: Thu 2/6/25 Dur: 30 days
Comp: 0%
4.1 Resource Management
Start: 12/27/24 ID: 32
Finish: 1/6/25 Dur: 7 days
Res: Resource Manager, Project Manager
4.2 Cost Management
Start: 1/7/25 ID: 33
Finish: 1/16/25 Dur: 8 days
Res: Finance Department, Project Manager
5.0 Closing
Start: Fri 2/7/25 ID: 36
Finish: Fri 3/21/25 Dur: 31 days
Comp: 0%
5.1 Prepare final project report
Start: 2/7/25 ID: 37
Finish: 2/24/25 Dur: 12 days
Res: Project Manager
4.3 Progress reports
Start: 1/17/25 ID: 34
Finish: 1/28/25 Dur: 8 days
Res: Project Manager, Project Team
4.4 Change requests
Start: 1/29/25 ID: 35
Finish: 2/6/25 Dur: 7 days
Res: Change Management Specialist, Project Manager
5.2 Prepare final project presentation
Start: 2/25/25 ID: 38
Finish: 3/3/25 Dur: 5 days
Res: Project Manager, Project Team
5.3 Lessons learned
Start: 3/4/25 ID: 39
Finish: 3/21/25 Dur: 14 days
Res: Project Manager, Project Team

Activity Cost Estimation

Cost estimate

XXX Project Cost Estimate

Prepared by: **Date:**

Note: Change the WBS items and other entries to meet your project needs. This data is from Figure 7-1 of Schwalbe's text Information Technology Project Management, Fourth Edition. Also make sure the formulas work properly based on the data you enter.

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 1 Totals	% of Total
WBS Items					
1. Project Management				**$15,000**	**5%**
1.1 Project manager	40	$75	$3,000		
1.2 Determine which companies	80	$100	$8,000		
1.3 Prepare project charter	20	$75	$1,500		
1.4 Hold project kick-off meeting	10	$75	$750		
2.0 Planning				$25,000	**8%**
2.1 Hold team planning meeting	40	$75	$3,000		
2.1.1 Determine the responsibilities of each team member	20	$75	$1,500		
2.2 Prepare scope statement	30	$75	$2,250		
2.3 Prepare WBS	20	$75	$1,500		
2.4 Prepare schedule and cost baseline	80	$75	$6,000		
2.4.1 Preparing the project start and end schedule	20	$75	$1,500		
2.4.2 Explain the steps of the project	20	$75	$1,500		
2.4.3 Determine the duration of each step	20	$75	$1,500		
2.4.4 Create draft Gantt char	10	$75	$750		
2.4.5 Clarify the programs used and their cost	10	$75	$750		
3.0 Executing				**$163,489**	**53%**
3.1 Procurement	400	$102	$40,872		
3.1.1 Procure RFID readers and tags/cards			$13,624		
3.1.2 Procure Fingerprint recognition devices	60	$227	$13,624		
3.1.3 Procure Cameras	60	$227	$13,624		
3.2 Development			$40,872		
3.2.1 Develop Attendance management software	60	$227	$13,624		
3.2.2 Integrate with existing systems	60	$227	$13,624		
3.3 Training			$40,872		
3.3.1 Conduct User Training	60	$227	$13,624		
3.3.2 Implement Change Management Initiatives	60	$227	$13,624		
3.4 Testing			$40,872		
3.4.1 Conduct System Testing			$40,872		
4.0 Monitoring and Controlling			$30,000	**$30,000**	**10%**
4.1 Resource Management	100	$75	$7,500		
4.2 Cost Management	80	$75	$6,000		
4.3 Progress reports	60	$75	$4,500		
4.4 Change requests	40	$75	$3,000		
5.0 Closing			$12,555	$12,555	4.00%
5.1 Prepare final project report			$3,750		
5.2 Prepare final project presentation			$1,500		
5.3 Lessons learned:			$2,250		
6. Reserves (20% of total estimate)			$61,511	**$61,511**	**20%**
Total project cost estimate				**$307,555**	100%
* See software development estimate					